

Electrical Characterization organic FETs and Comparison with ALD1106

May 11th, 2026

1. Electrical characterization of ALD1106 and PS9 using transfer/output characteristics.
2. Results of C-gvdP method:
ALD1106: $\mu = (1267 \pm 3) \text{ cm}^2/(\text{V s})$, $R_{\infty} = (88.1 \pm 0.2) \Omega$;
PS9: $\mu = (3.29 \pm 0.03) \text{ cm}^2/(\text{V s})$, $R_{\infty} = (4.3 \pm 0.1) \text{ M}\Omega$.
3. Illumination has no noticable effect.
4. DLTS method indicates charge trapping and capacitance of $C \approx 2.00 \text{ nF}$.
5. Small-signal amplification properties of PS9.

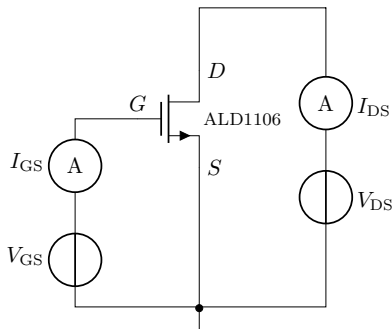


Figure: Circuit layout for I-V characterization.

$$I_{DS} = \begin{cases} 0, & \text{Cut-off, } V_{GS} < V_{th} \\ \mu C_i \frac{W}{L} \left[(V_{GS} - V_{th}) V_{DS} - \frac{V_{DS}^2}{2} \right], & \text{Linear, } V_{GS} > V_{th}, V_{DS} < V_{GS} - V_{th} \\ \frac{1}{2} \mu C_i \frac{W}{L} (V_{GS} - V_{th})^2, & \text{Saturated, } V_{GS} > V_{th}, V_{DS} \geq V_{GS} - V_{th} \end{cases}$$

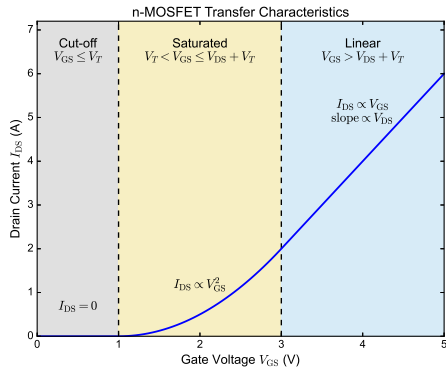
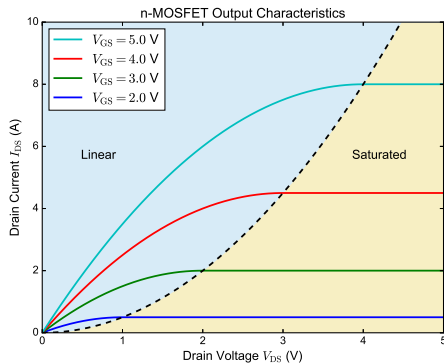


Figure: Theoretical I-V Characteristics of n-MOSFET (Simulated by Python).

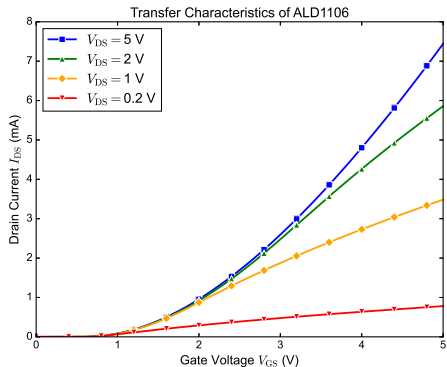
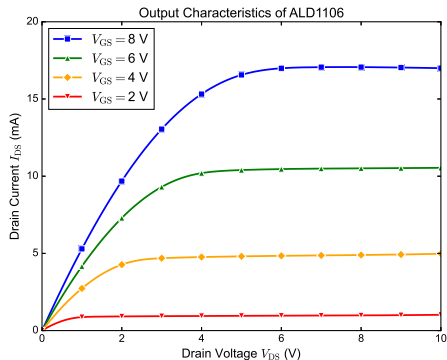


Figure: I-V Characteristics of ALD1106.

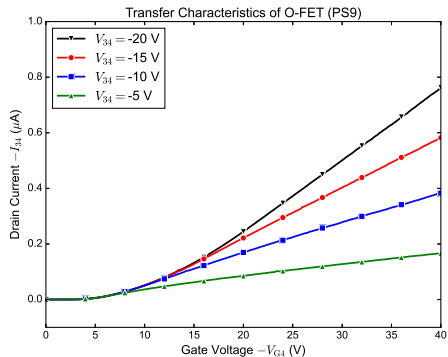
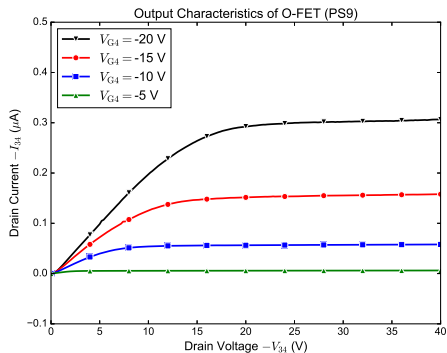


Figure: I-V Characteristics of Sample PS9.

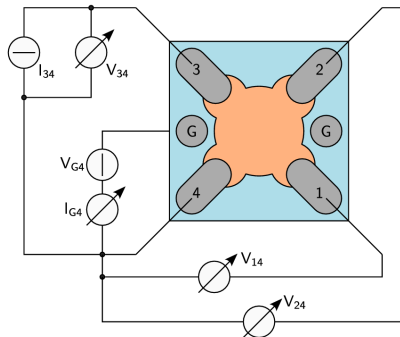


Figure: Circuit layout for vdP measurements (Source: [RK-Figures](#)).
Equations for generic TFT model and vdP configuration:

$$I_{34} \frac{L}{W} = \mu C_i \left(V_{G4} - V_T - \frac{V_3 + V_4}{2} \right) V_{34} \equiv \sigma V_{34},$$

$$I_{34} \frac{\ln 2}{\pi} \equiv \sigma V_{21} = \mu C_i \left(V_{G4} - V_T - \frac{V_1 + V_2}{2} \right) V_{21}.$$

We set contact 4 as ground, so $V_4 = 0$.

The conductivity σ can be written in two equivalent forms:

$$\sigma = \mu C_i \left(V_{G4} - V_T - \frac{V_3}{2} \right),$$
$$\sigma = \mu C_i \left(V_{G4} - V_T - \frac{V_1 + V_2}{2} \right) \quad (\text{vdP}).$$

Because V_{34} also contains the contact-resistance drop, V_3 cannot be measured directly.

The vdP method avoids this issue by using V_1 and V_2 instead.

We therefore define $V_C = \frac{V_1 + V_2}{2}$.

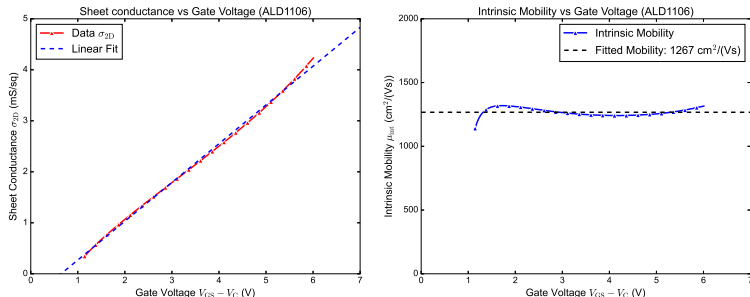


Figure: Extraction of threshold voltage, charge-carrier mobility of ALD1106.

Fit results: $\sigma_{2D} = \mu C_i (V_{GS} - V_C - V_T) \rightarrow y = ax + b$

(whereas $y = \sigma$, $x = V_{GS} - V_C$, $V_C = (V_1 + V_2)/2$, $C = 3 \text{ pF}$, $A = 500 \text{ } \mu\text{m}^2$, $C_i = C/A$).

$$\mu = \frac{a}{C_i} \cdot 10^4 = (1267 \pm 3) \text{ cm}^2/(\text{Vs}),$$

$$V_T = -\frac{b}{a} = (0.644 \pm 0.009) \text{ V},$$

$$r^2 = 0.998.$$

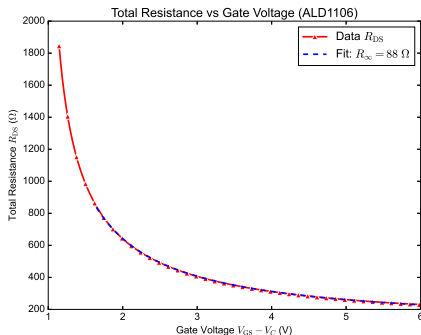


Figure: Extraction of background resistance of ALD1106.

Fit results: $R_{DS} = \frac{V_{DS}}{I_{DS}} = R_{\infty} + \frac{b}{V_{GS} - V_C - V_T} \rightarrow y = a + \frac{b}{x}.$

$$R_{\infty} = (88.1 \pm 0.2) \, \Omega,$$

$$r^2 = 0.99993.$$

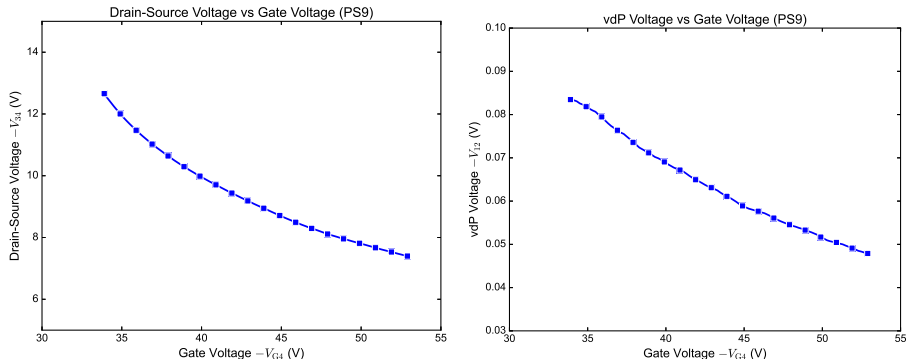


Figure: Voltage of contacts in vdP measurement.

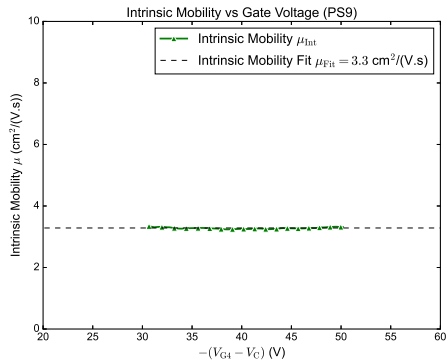
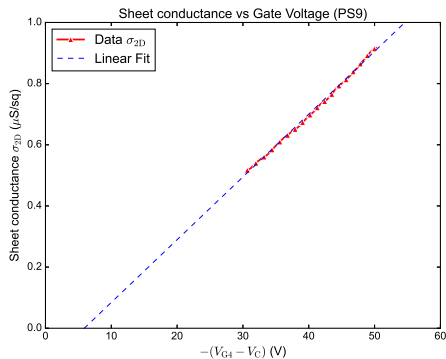


Figure: Extraction of V_T , μ of PS9

Fit results:

$$\mu = (3.29 \pm 0.03) \text{ cm}^2/(\text{V}\cdot\text{s}),$$

$$V_T = (-5.9 \pm 0.4) \text{ V},$$

$$r^2 = 0.991.$$

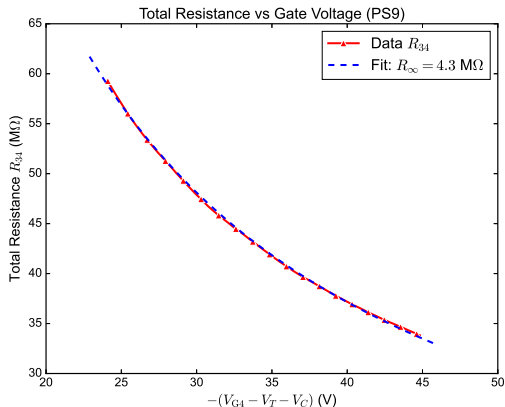


Figure: Extraction of background resistance of PS9 ($I_{34} = 300$ nA).

Fit results:

$$R_{\infty} = (4.3 \pm 0.1) M\Omega,$$

$$r^2 = 0.9992.$$

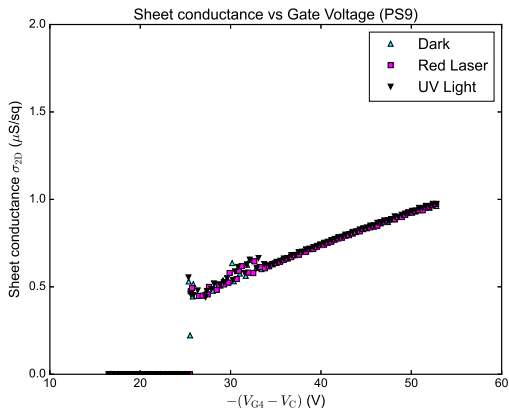


Figure: Effects of illumination to the measurements.

Discussion: Illumination has no noticeable effect on the measured data.

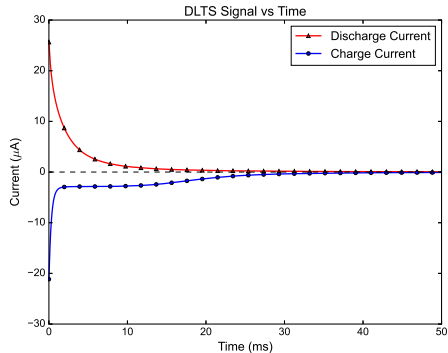
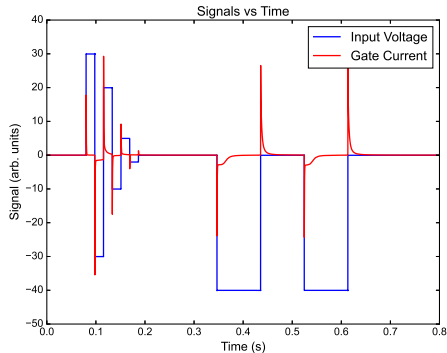


Figure: DLTS analysis for PS9.

1. Charging process:

$$|Q_1| = \int |I_1| dt = \sum_i |I_1(i)|(t_{i+1} - t_i) = 6.80 \cdot 10^{-8} \text{ C},$$
$$C_1 = \frac{|Q_1|}{V_0 - V_T} = 2.00 \text{ nF}.$$

2. Discharging process:

$$|Q_2| = \int |I_2| dt = \sum_i |I_2(i)|(t_{i+1} - t_i) = 6.44 \cdot 10^{-8} \text{ C},$$
$$C_2 = \frac{|Q_2|}{V_0 - V_T} = 1.89 \text{ nF}.$$

3. Trapped charges:

$$|\Delta Q| = |Q_1| - |Q_2| = 0.36 \cdot 10^{-8} \text{ C} \approx 5.3\% |Q_1|.$$

After the pulse, trapped charges either de-trap as a small relaxation current or recombine inside the device. If the waiting time is too short, residual trapped charges can cause memory effects, baseline drift, and affect the next pulse.

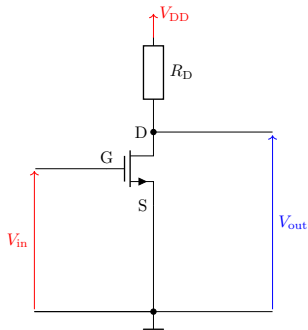


Figure: Circuit for simple Amplifier.

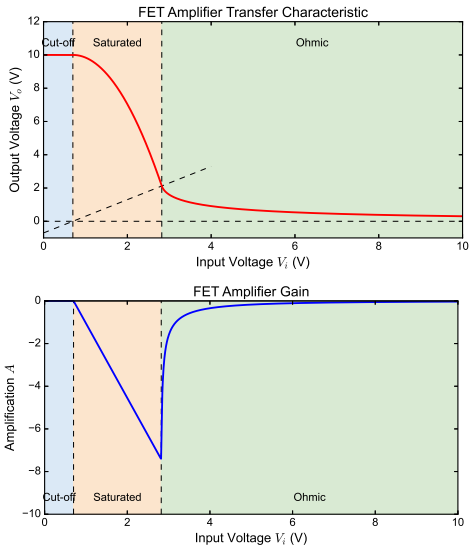


Figure: Transfer and Amplification.

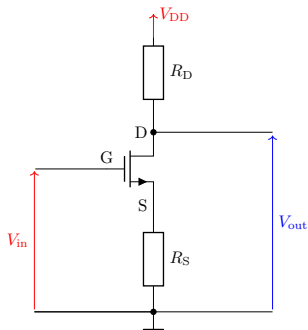


Figure: Circuit for Amplifier with Stabilizer resistor.

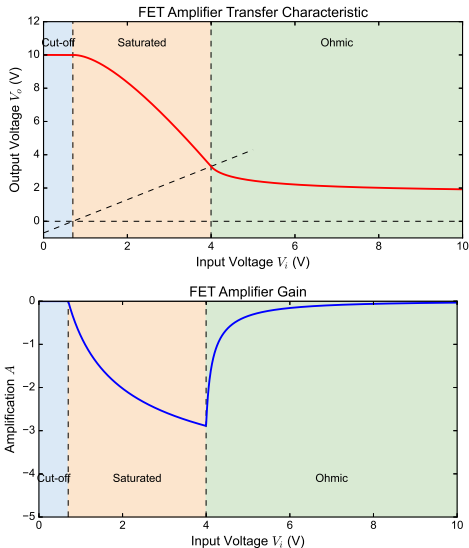


Figure: Transfer and Amplification.

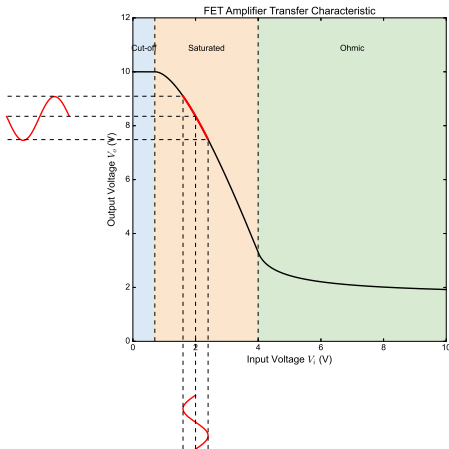


Figure: FET as Small-Signal Amplifier (Simulated by Python).

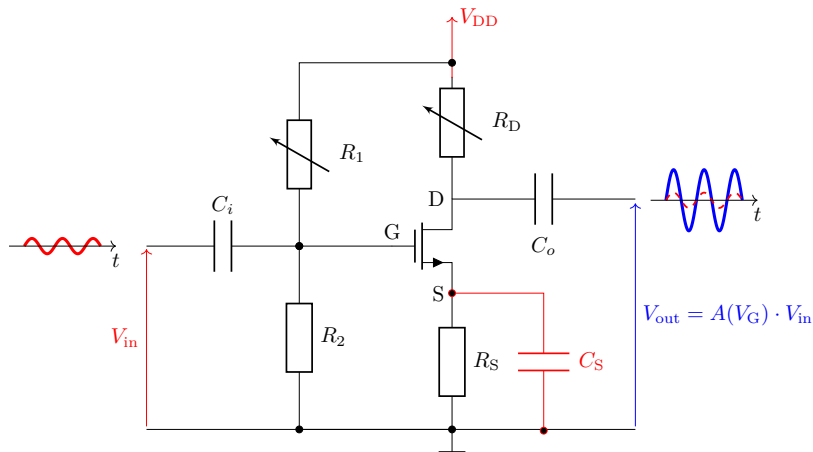


Figure: Class A FET Small-Signal Amplifier Circuit (For organic FETs, an external op-amp buffer or high impedance probe (100 M Ω) at output for measuring).

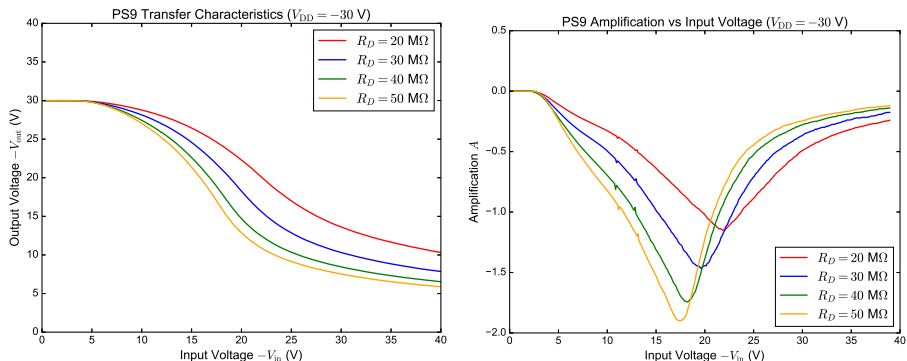


Figure: Transfer and Amplification Characteristics.

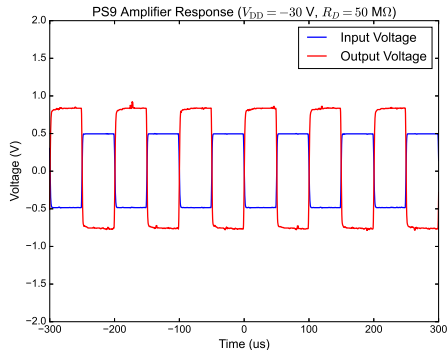
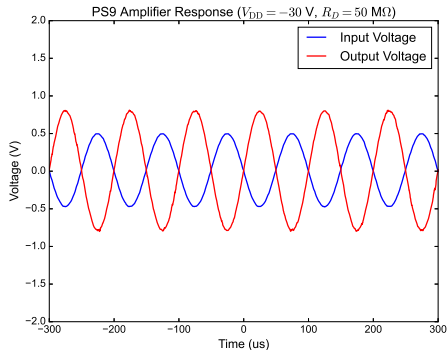
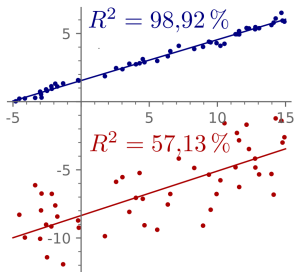


Figure: Signal Amplification of PS9 ($f = 10\text{ kHz}$).



- The sum of squares of residuals, also called the **residual sum of squares**:

$$SS_{\text{res}} = \sum_i (y_i - f_i)^2 = \sum_i e_i^2$$

- The **total sum of squares** (proportional to the **variance** of the data):

$$SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2$$

The most general definition of the coefficient of determination is

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

```
# Coefficient of determination
ss_res = np.sum((R_DS - R_fit) ** 2)
ss_tot = np.sum((R_DS - np.mean(R_DS)) ** 2)
r2 = 1 - (ss_res / ss_tot)
```

Figure: Top: Definition of r^2 (Source: [Wikipedia](#)). Bottom: Calculation of r^2 in Python.

Thank you for your attention!
Any questions?